Phenotypic Evolution: Process

Population Genetics

Evolution is a change in the characteristics of a population of organisms,

---but behind that change in characteristics is genetic change.

In population genetic terms, evolution is defined as a change in the frequency of specific gene in a population.

ABO blood system as an example

The frequency of the gene that codes for A blood protein, I^A

I^A =the number of copies of that gene divided by the total number of all A, B or O blood protein coding genes in the population.

If I^A = 26.1 % then,
I^B, I^0 blood coding genes made up 13.4% and 60.5 % of the blood protein coding genes
Total adding up to 100%

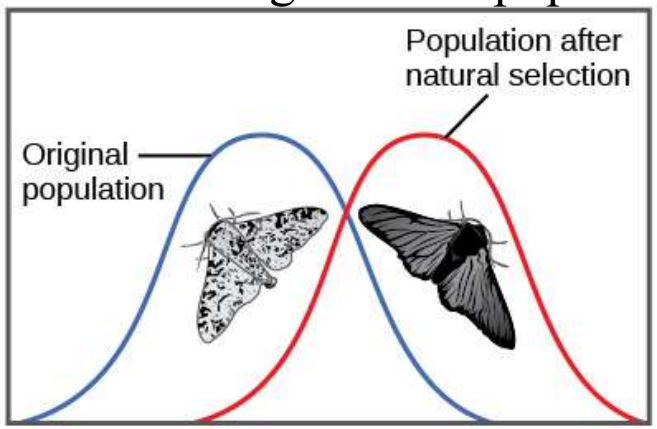
A change in this frequency over time would constitute evolution in the population.

How can this change happen?

One way the frequency of a particular gene in a population can change is **natural selection.**

Highly beneficial genes may, over a very few generations, become "fixed" in this way, meaning that every individual of the population will carry the gene.

Similarly, detrimental genes may be swiftly eliminated from the **gene pool** (the sum of all the genes in a population).



Light-colored peppered moths are better camouflaged against a pristine environment; likewise, dark-colored peppered moths are better camouflaged against a sooty environment. Thus, as the Industrial Revolution progressed in nineteenth-century England, the color of the moth population shifted from light to dark.

Part of the study of population genetics is tracking how selective forces change the frequencies of certain genes in a population over time, which gives us clues about the selective forces that may be operating on a given population.

Mechanisms of Evolution

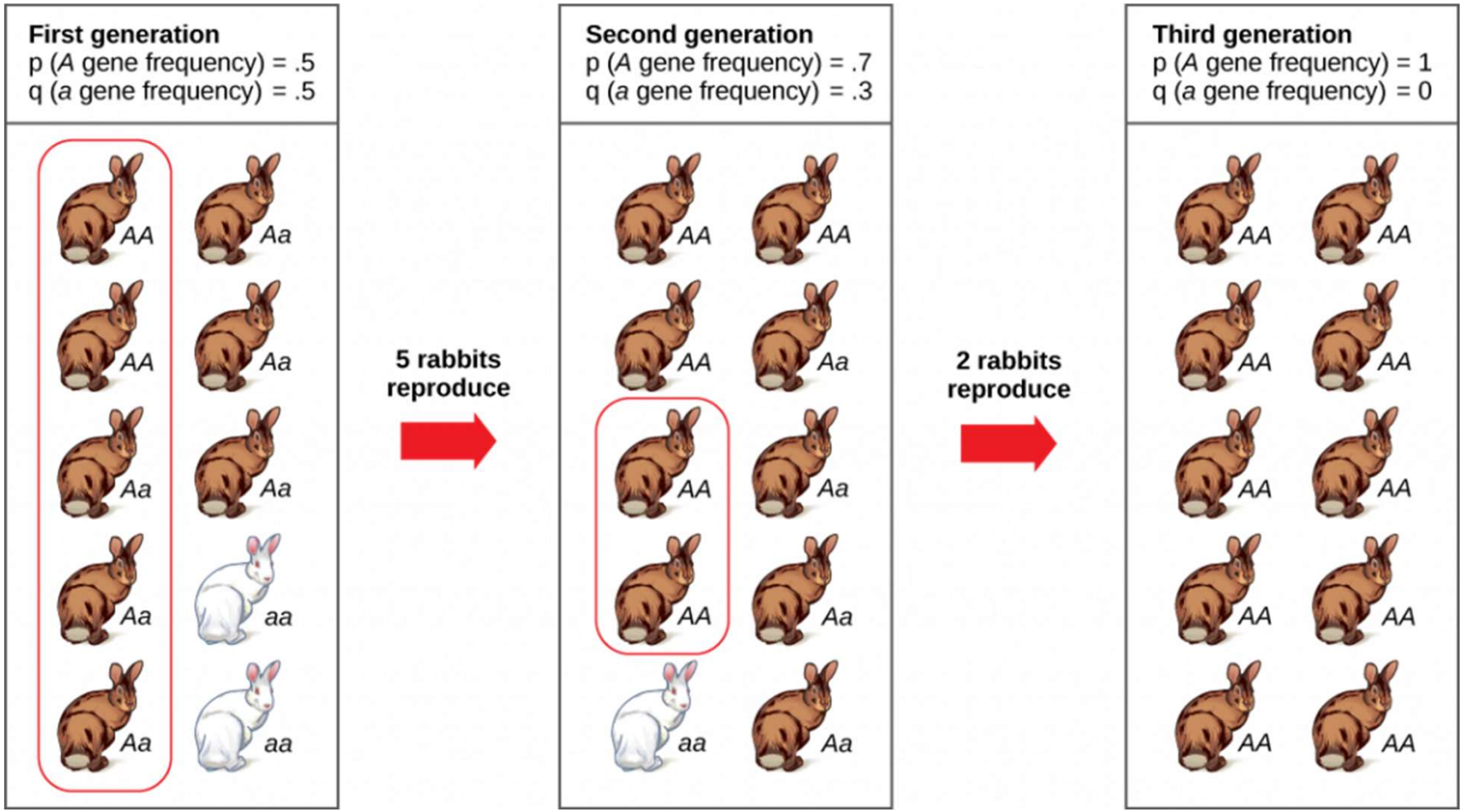
- **Natural Selection**
- **Mutation**

-- change in the DNA sequence of the gene….. Effect relatively small, unless **it acts with Natural selection**

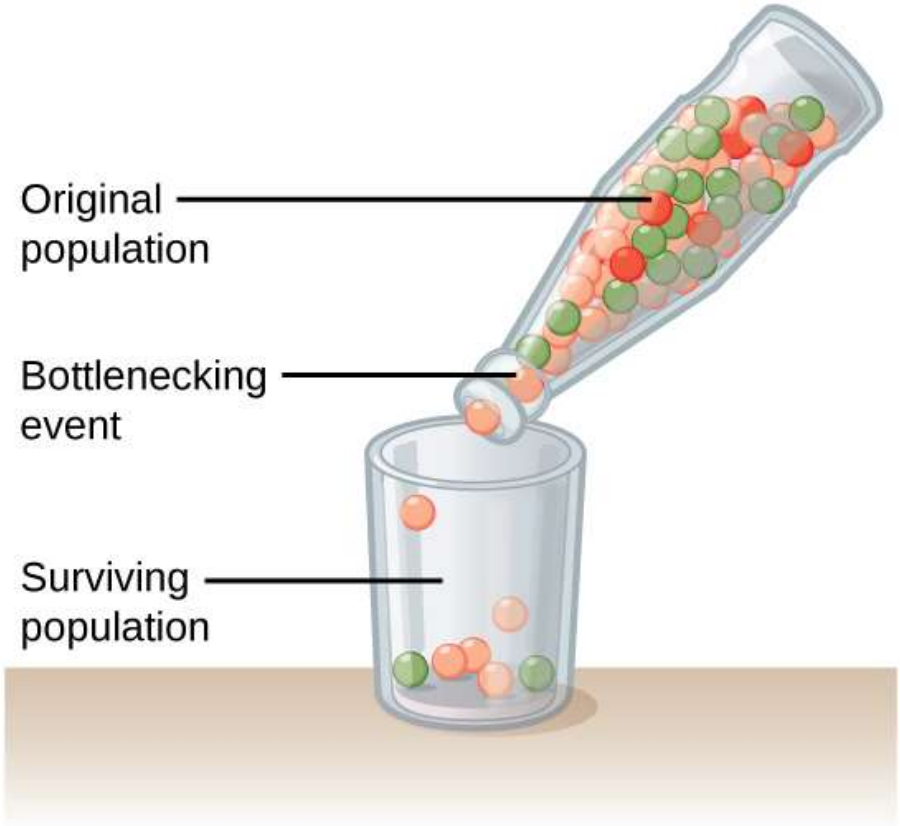
-- It should be noted that mutation is the ultimate source of genetic variation in all populations—new alleles, and, therefore, new genetic variations arise through mutation.

-- A mutation may produce an allele that is selected against, selected for, or selectively neutral.

- **Genetic Drift**



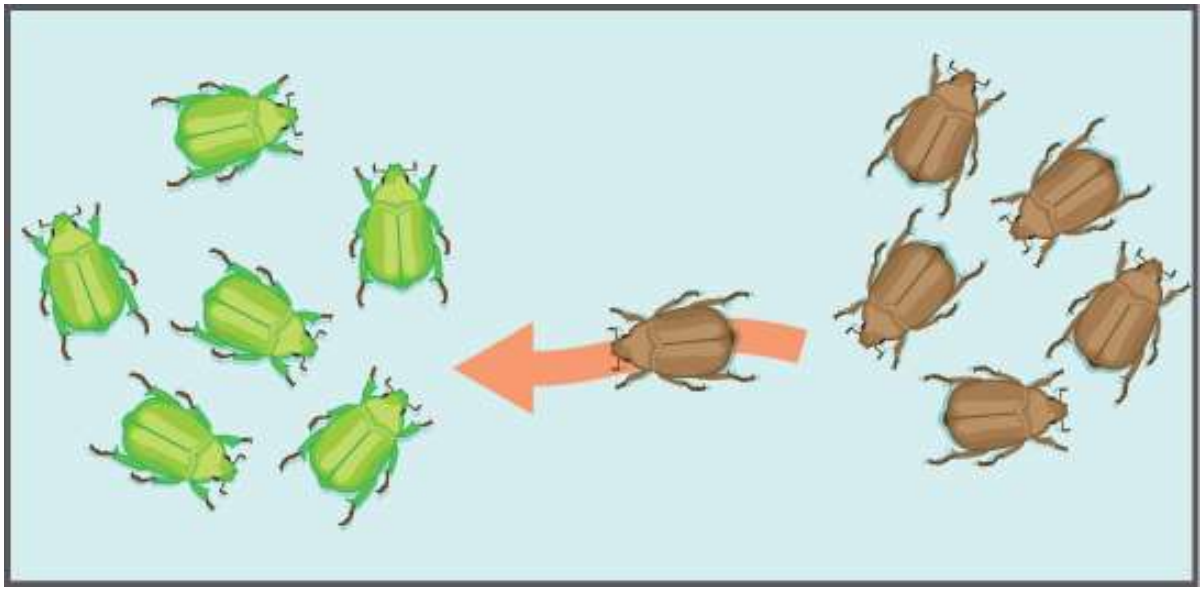
Genetic drift can also be magnified by natural or human-caused events, such as a disaster that randomly kills a large portion of the population, which is known as the **bottleneck effect** that results in a large portion of the gene pool suddenly being wiped out



Another scenario in which populations might experience a strong influence of genetic drift is if some portion of the population leaves to start a new population in a new location, or if a population gets divided by a physical barrier of some kind.

- **Gene Flow**

Another important evolutionary force is **gene flow**, or the flow of genes in and out of a population resulting from the migration of individuals or gametes



Gene flow can occur when an individual travels from one geographic location to another and joins a different population of the species. In the example shown here, the brown allele is introduced into the green population.

Many plants, for example, send their seeds far and wide, by wind or in the guts of animals

Genetic diversity can be lost through:

- Drift : change in the frequency of a gene variant (allele) in a population due to random sampling
- Bottlenecks
- Founder effects : loss of genetic variation that occurs when a new population is established by a very small number of individuals from a larger population.
- Inbreeding
- All of these are closely related

Random Events

There is a random element involved in the frequency of genes in one generation when compared to the previous generation.

Genetic Drift: a random change in the genetic makeup of a population that may arise when the population is restricted in size.

"Drift" is directionless.

Compare to tossing a coin.

In Florida, the greenhouse frog *Eleutherodactylus planirostris* occurs in two pigment patterns, mottled and striped. Why?

The theory of genetic drift was developed in the 1930s by Sewall Wright, and is sometimes referred to as the **Sewall Wright effect**.

The size of a population affects the rate at which a mutation can spread.

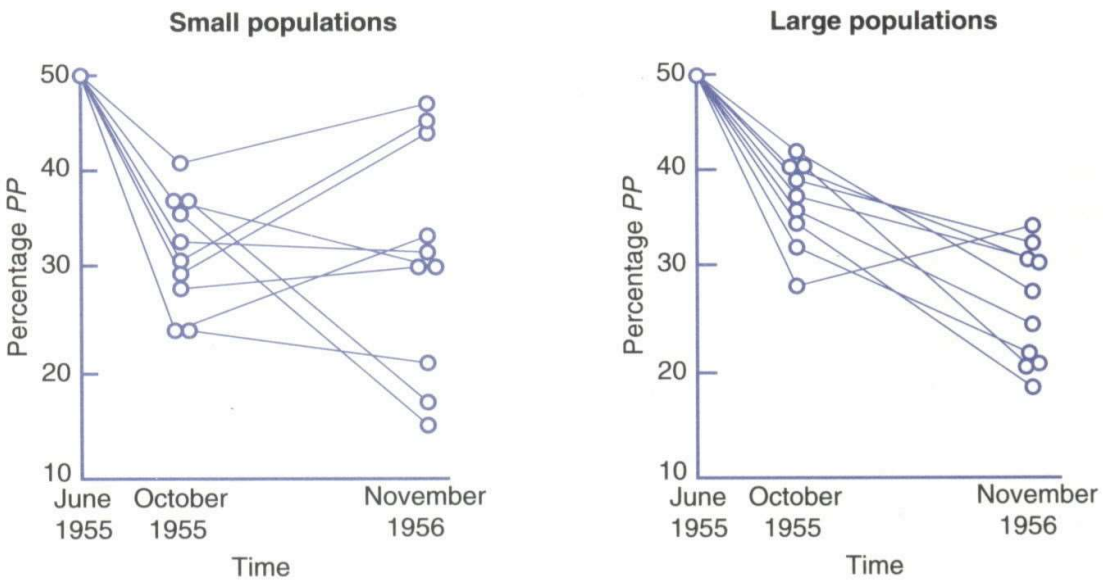
TABLE 9.1 | Chance Distribution of Offspring of Two Heterozygous Parents (Aa × Aa)

Genotype of first offspring	Probability of first event	Genotype of second offspring	Probability of second event	Total Probability
AA	1/4	*AA*	1/4	Both offspring *AA*, 1/16
AA	1/4	*Aa*	2/4	*AA* followed by *Aa*, 2/16
AA	1/4	*aa*	1/4	*AA* followed by *aa*, 1/16
Aa	2/4	*AA*	1/4	*Aa* followed by *AA*, 2/16
Aa	2/4	*Aa*	2/4	Both offspring *Aa*, 4/16
Aa	2/4	*aa*	1/4	*Aa* followed by *aa*, 2/16
aa	1/4	*AA*	1/4	*aa* followed by *AA*, 1/16
aa	1/4	*Aa*	2/4	*aa* followed by *Aa*, 2/16
aa	1/4	*aa*	1/4	Both offspring *aa*, 1/16

Consider a mathematical example. If two Aa frogs have only two offspring, there is a 1/16 chance of the a gene (or the A gene, for that matter) being lost after the first generation.

The rate of change of gene frequency by random drift depends on the size of the population.

Pop size= 20 individuals



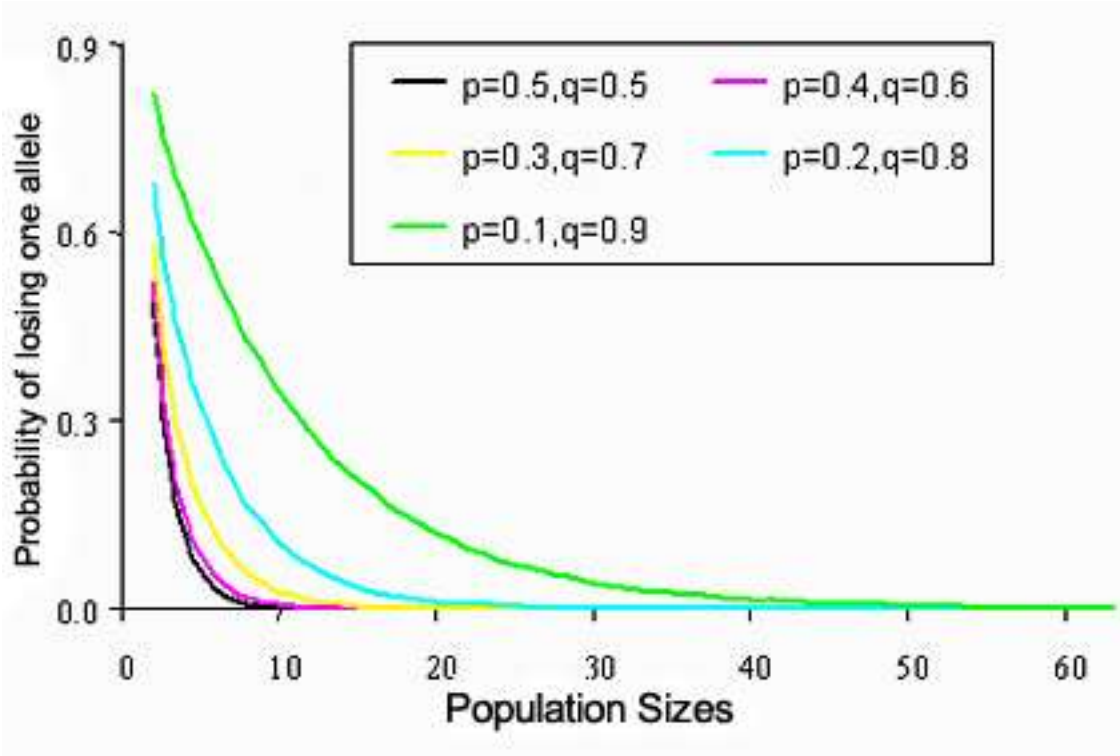
Pop size= 4000 individuals

Random sampling is more effective in small populations.

Random sampling is more effective at creating variations in small populations than in large ones. Results from the Dobzhansky experiment with Drosophila populations.

- **FOUNDER EFFECT**

The establishment of a new population by a few original founders (in an extreme case, by a single fertilized female) which carry only a small fraction of the total genetic vari



Founder effect: unique gene frequencies arising in small groups of individuals that migrate away from a large population.

Figure: The chance that a founder population will be homozygous depends on the number of founders and the gene frequencies.

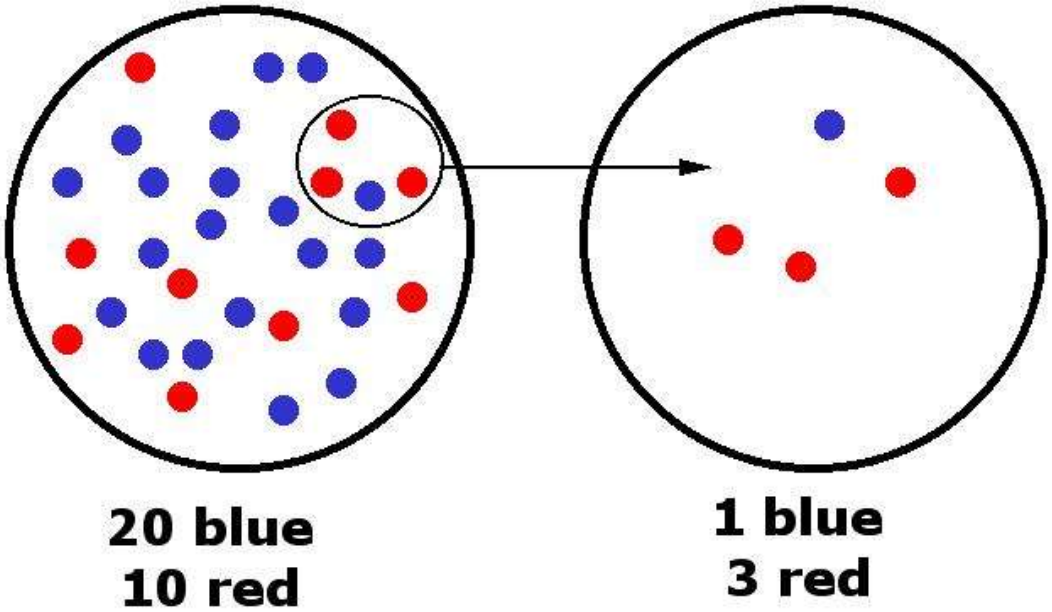
A population may be descended from a small number of ancestral individuals for two reasons:

REASON One

• A small number of individuals may colonize a place previously uninhabited by their species.

E.g. The 250 people making up the human population on the island of Tristan da Cunha, are mostly descended from one Scottish family who arrived in 1817.

Founder effect



The ~300 inhabitants of Tristan da Cunha are descended from 20 or so 19th Century immigrants.

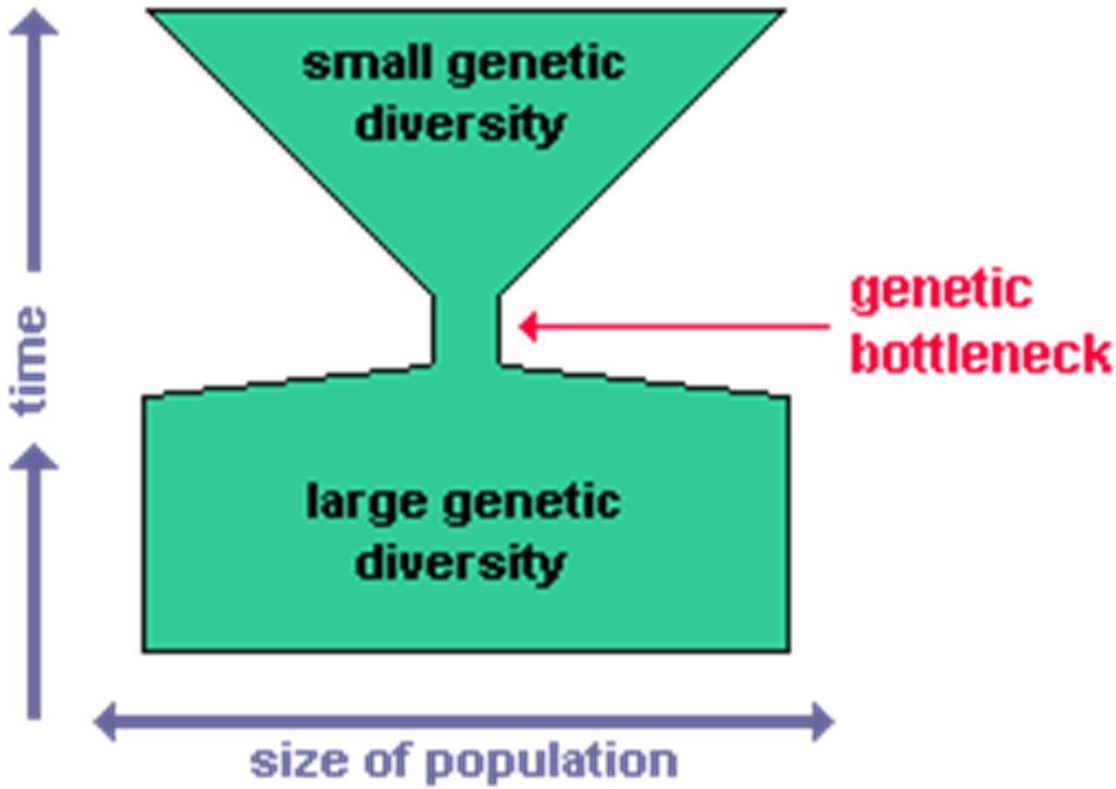
Native Americans provide a possible example. Most North American tribes lack the gene governing type B blood. However, that gene is widespread in Mongolia, their ancestral home. The group that migrated across the Bering Strait may have been small, and lacked this gene.

Native American populations also show a high incidence of albinism. Can this also be attributed to founder effects?

REASON Two

- An established population may fluctuate in size: the founder effect occurs when the population passes through a 'bottleneck' in which only a few individuals survive, and later expands again under more favorable conditions.

Thus, a large, established population is reduced in size. The remaining individuals may not be representative of the genetic diversity that was present in the original population. This is referred to as a genetic bottleneck.



If a population passes through a “bottleneck”, the amount of genetic diversity may be greatly lowered.

Possible consequences of genetic drift in small populations?

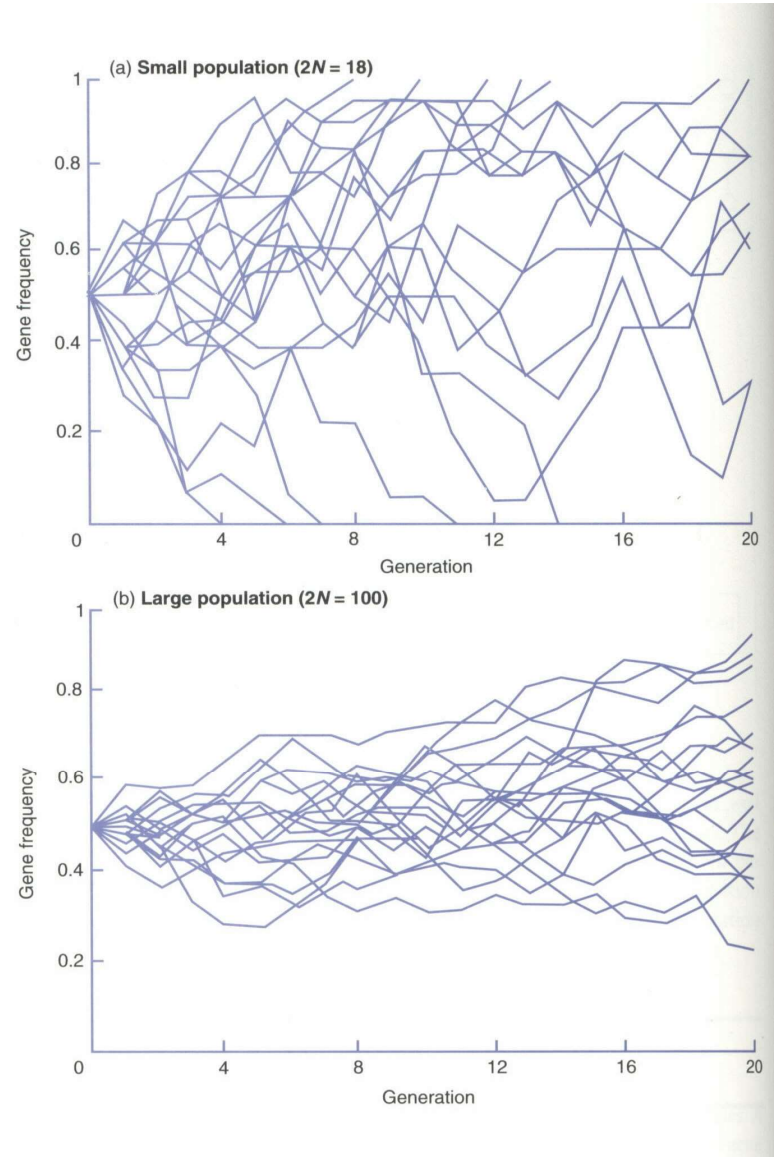
The drift to homozygosity- Over time, the population must drift to become made up of the descendants of only one gene from an ancestral population. Any one gene has a 1/2N chance of eventual fixation by random drift.

Small populations will lead to increase in homozygosity, decrease in heterozygosity, and increased genetic differences between populations

Variation in sex ratio, population size, will lead to inbreeding

In small populations, Hardy-Weinberg equilibrium is not reached until they reach homozygosity.

Homozygosity is the result in any small population without mutation.



- Small populations have less genetic variation than large ones
- The relationship is non-linear
- Depends on **effective** population size (Sewall Wright)

> *The **effective population size** is the **size** of an ideal **population** (i.e., one that meets all the Hardy-Weinberg assumptions) that would lose heterozygosity at a rate equal to that of the observed **population**.*

More generally, an effective population size may be defined as the number of individuals in an idealised population that has a value of any given population genetic quantity that is equal to the value of that quantity in the population of interest.

Effective population size

It is the equivalent population with

- Random mating
- Non-overlapping generations
- Even distribution of births and deaths
- No selection
- In general effective population size<actual population size

When it varies from 1:1 sex ratio

$$4N_m \bullet N_f /(N_m + N_f)$$

e.g with 80 males and 20 females (an absolute population size of 100)

$$N_e = \frac{4 \times 80 \times 20}{80 + 20} = 64$$

Populations lose genetic variability over time through chance effects = drift

- Rate of drift is proportional to $1/N_e$

- Small populations lose diversity faster than large ones
- Rare alleles are more likely to be lost than common ones

- If effective population size is small, relatives are more likely to mate with each other and deleterious recessives are more likely to be expressed
- = **inbreeding depression**
- If a bottleneck is large enough and short enough, it can purge population of deleterious recessives

- Historically, it was thought that genetic stochasticity was less important than other forms of stochasticity
- May not be the case
- **Mutational meltdown**
- **Extinction vortex** (Michael Soulé)

Small population size leads to

- Increased inbreeding
- Reduced heterozygosity
- Expression of semi-lethal recessives
- Reduced fecundity and survival
- Further reduction in population size
- So far, rather little empirical evidence that this occurs in nature